Rakib - AI Helmet for Safety

Student Name 1: Layla AlMaseeh

Student Name 2: Reem Al-Sulaiman

Student Name 3: Moza Al-Naimi

Student Name 4: Nadeen Abunada

FUNCTIONAL SYSTEM REQUIREMENTS

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Approved by:

Student Name 1: Layla AlMaseeh 24/10/25

Student Name 2: Reem Al-Sulaiman 24/10/25

Student Name 3: Moza Al-Naimi 24/10/25

Student Name 4: Nadeen Abunada 24/10/25

Course Instructor: Dr. Ali Ghrayeb 24/10/25

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1. Introduction

1.1 Purpose and Scope

The Rakib Helmet is designed to improve the safety of motorcycle delivery riders by detecting both fatigue and crash events in real time This system provides early prevention and rapid emergency response, addressing the high accident risks faced by delivery riders. The project's scope is to develop a fully functional prototype of the Rakib Helmet capable of operating for several hours. This prototype will accurately detect fatigue and crash events and initiate communication alerts, through the use of artificial intelligence. The figure below shows the project diagram in the proposed CONOPS and an explanation of the data flow between the components.

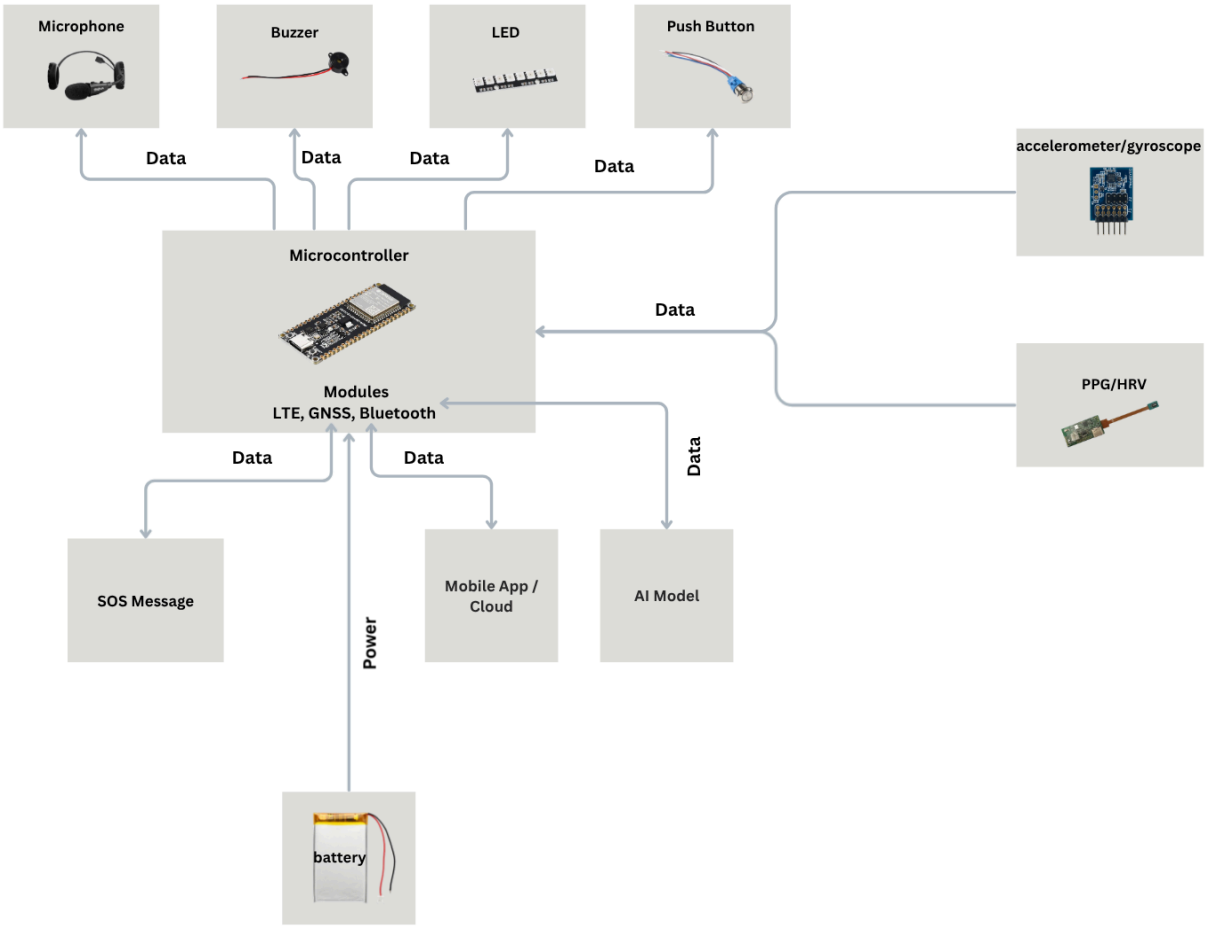

***Figure 1**: Functional System Diagram of Rakib Helmet*

The system is going to be powered using a lithium battery that will keep the system working for several hours. The main brain of the system is the LilyGO T-SIM7600G-S3 (ESP32) microcontroller that has a built-in bluetooth and LTE/GNSS module. The helmet tracks heart rate. Oxygen levels, and impact force using two sensors, the MAXM86161EVSYS (PPG/HRV) sensor and the MPU-6050 (Accelerometer/gyroscope) sensor. The data collected from the sensors is processed by the ESP32. Then transmitted to the AI model where it will be analyzed to identify signs of fatigue or crash events.

As shown in figure 1 there will be a two way data connection between the communication module and the SOS message, the AI model and the mobile app/cloud. This will ensure accurate data exchange with alerts and post ride analytics then the microcontroller turns on the led, buzzer, and microphone for visual and sound alerts as soon as signs of fatigue are detected, such as irregular heartbeats and oxygen levels. The system will send an SOS message with GPS coordinates to emergency services in the case of a crash using the LTE/GNSS module.

1.2 Responsibility and Change Authority

The project team is responsible for ensuring that all specifications and functional requirements of the Rakib Helmet system are met. Any changes to the project's specifications or deliverables must be reviewed and approved by the team leader Layla Al Maseeh and supervising mentor, Dr. Ali Ghrayeb.

2. Applicable and Reference Documents

2.1 Applicable Documents

The following standards and specifications, performance, and safety criteria are to be adopted during the development of the Rakib AI Helmet for Safety & Fatigue Monitoring system. Applicable documents are listed in Table 1.

Document Number / Source	Revision / Release Date	Document Title
IEEE 1028	2008	IEEE Standard for Software Reviews and Audits
ISO 26262	2018	Road Vehicles Functional Safety Standard
ISO 45001	2018	International Standard for Occupational Health and Safety Management Systems
IEEE 802.15.1	2017	IEEE Standard for Bluetooth Wireless Communication
ASTM F1446	2025	Standard Test Methods for Equipment and Procedures

ISO/IEC 27001	2023	International Standard for Information Security Management Systems for protecting user and cloud data.
IEC 60529	2020	Degrees of Protection Provided by Enclosures (IP Code)

Table 1: Applicable documents

2.2 Reference Documents

The documents that are used as reference to conceptualize and to design the Rakib helmet are listed in Table 2 These references support the research, design methodologies, and system modeling efforts. They do not form a controlled part of this specification but provide useful guidance.

Document Number / Source	Revision / Release Date	Document Title
WHO Road Safety Report	2023	Global Status Report on Road Safety
IEEE Xplore	2024	Research Articles on AI-Based Driver Fatigue Detection
TensorFlow Developer Guide	2024	Deep Learning Framework for Real-Time Analysis
LilyGo	2024	T-SIM7600G-S3 (ESP32-S3) Hardware Reference and Pinout Guide
SIMCom	2023	SIM7600G Hardware Design and AT Command Manual
Firebase Documentation	2024	Firebase Real-Time Database and Cloud Messaging API Reference
MAXM86161 Datasheet (Maxim Integrated)	2023	PPG/HRV Sensor for Heart Rate and SpO2 Monitoring
MPU-6050 Datasheet (TDK InvenSense)	2023	6-Axis Gyroscope and Accelerometer for Motion Tracking
NHTSA Helmet Safety Guidelines	2022	Safety Requirements and Testing Standards for Protective Headgear

Table 2: Reference documents

2.3 Order of Precedence

In case of conflicts between this specification and the references, the contents of this specification should take complete precedence. All external publications serve as guidance and are not mandatory unless explicitly incorporated into this document.

3. Requirements

In this section the term “Rakib AI Helmet for Safety" refers to the full system being developed as a proof of concept which includes the helmet’s built in sensors, main microcontroller, power and alert components, communication modules, and the online database where all data is stored and displayed.

3.1 System Definition

Six subsystems are shown in figure 2 which work together to make up the Rakib helmet: Sensor, Microcontroller, Communication, Power, Alert and User Interface, and Information Storage and Analysis . Each part plays a specific role. Together they create a complete safety system that is capable of detecting crashes, rider fatigue, and transmitting critical information to the external systems for real-time response and this system is designed to collect and share data efficiently while maintaining rider stability, comfort, and safety standards.

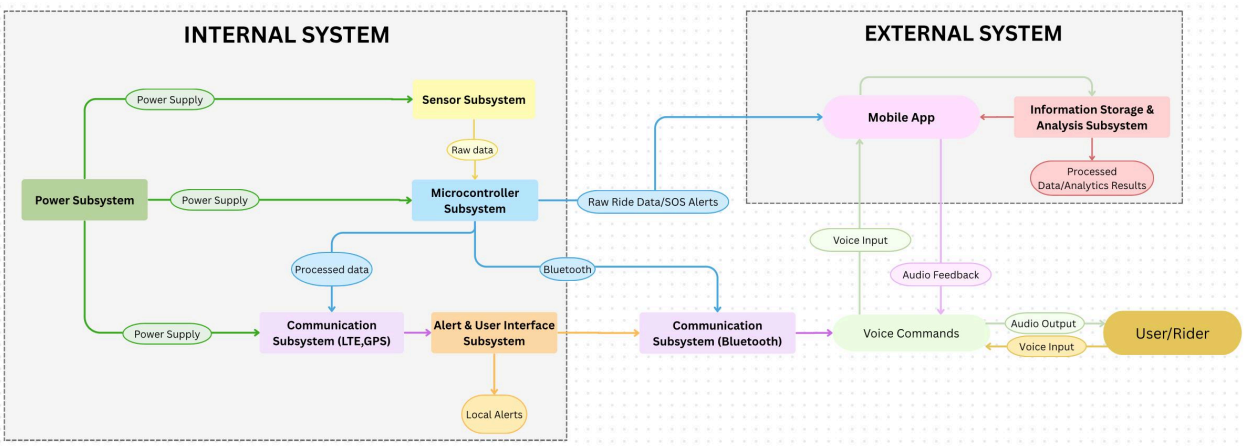

Figure 2: Block diagram of Rakib’s internal and external system

Power Subsystem
This subsystem provides stable power to all electronic components and uses a rechargeable lithium-polymer (Li-Po) battery and is managed by a built-in Battery Management System (BMS) to prevent overcharging, short-circuit conditions, and maintain battery efficiency [6]. Voltage regulators help maintain the supply for the sensors and microcontroller and the system operates for about eight hours on a single charge and distributes power to the Sensor, Microcontroller, Communication, and Alert & User Interface subsystems.

Sensor Subsystem
The sensor subsystem collects both body signals and motion data from the rider. It utilizes an accelerometer/gyroscope to measure acceleration, angular velocity, and impact forces, allowing

the detection of unusual head movement or crashes [23]. A PPG/HRV sensor measures the rider's heart rate, heart-rate variability, and oxygen levels to monitor fatigue or stress [22]. All raw data is transmitted from the sensors to the microcontroller for further processing.

Microcontroller Subsystem
The microcontroller subsystem serves as the main control and raw data collection unit of Rakib [19]. It is implemented using the ESP32 board, It receives raw data from the Sensor Subsystem and analyzes it to detect fatigue and crash conditions. Once processed, the microcontroller sends:

- Processed ride data and SOS alerts to the Communication Subsystem (LTE/GPS) for transmission to the mobile app and cloud [20].
- Bluetooth signals to the Communication Subsystem (Bluetooth) for local app connection and voice feedback [4].

It also forwards control signals to the Alert and User Interface Subsystem to activate LED and buzzer alerts.

Communication Subsystem
The communication subsystem manages all wireless connectivity necessary for operation and emergency response. There are two main communication paths:

1. **LTE/GPS Communication Subsystem:** Receives processed data and SOS alerts from the Microcontroller Subsystem and sends them to the external Mobile App and cloud database for location tracking and emergency response [20].
2. **Bluetooth Communication Subsystem:** Receives data and commands from the Microcontroller Subsystem and connects to the Mobile App for short-range communication [4]. It sends audio feedback and status updates to the User/Rider when the helmet is online, and relays voice commands from the app back to the helmet.

Alert and User Interface Subsystem
The Alert and User Interface Subsystem receives control signals from the Microcontroller Subsystem and provides real-time feedback to the rider. It includes LED indicators, a buzzer, a push button, and a speaker–microphone kit for communication through the Rakib mobile app. The LED shows the system status: blue means fatigue is detected, red means SOS is not sent, and green means SOS is sent. The buzzer emits a short audible countdown before the SOS is transmitted; allowing manual cancellation through the push button if necessary. The LED, buzzer, and push button operate offline to ensure that alerts and safety feedback are always available even without network connectivity.

Information Storage and Analysis Subsystem
This external subsystem includes the Mobile App / Cloud database. It receives ride data and SOS alerts from the helmet through the LTE Communication Subsystem and stores them for long-term analysis [9][18]. The Mobile App works with the AI to apply the analysis found and

updates the status by communicating the findings with the microcontroller via bluetooth to display ride history, fatigue alerts, and safety logs. [21]. These features help evaluate rider performance and improve fatigue detection accuracy over time.

3.2 Characteristics

3.2.1 Functional / Performance Requirements

3.2.1.1 Frequency of Measurements

- Accelerometer/gyroscope Sensor: Samples are taken at 100 Hz to record sudden acceleration or impact changes; Although 100 Hz provides sufficient temporal resolution for crash detection while consuming the least amount of power the device can support higher rates (up to 1 kHz) [23].
- PPG/HRV Sensor: samples at 50 Hz and analyses heart rate and HRV by internal averaging (4 samples → effective 12.5 Hz) [22].

Rationale: These sampling frequencies provide enough detail to detect immediate impacts and physiological variations in real time, while keeping processing and power demands balanced for a wearable system.

3.2.1.2 Accuracy of Measurements

Sensor	Parameter	Accuracy/Resolution
MPU-6050 Accelerometer sensor	Linear acceleration	±0.05 g (resolution ≈ 16-bit)
MPU-6050 Gyroscope sensor	Angular velocity	±0.2 °/s (resolution ≈ 16-bit)
PPG/HRV Sensor	Heart-rate / SpO_2	±2 BPM (typical), ±2% SpO_2
GPS ESP32	Position	±3 m (typical LTE-assisted)

Table 3: Required Accuracy of Sensors

Rationale: These accuracy levels are within high-end wearable standards. They are sufficient to distinguish actual crash events from normal movement and to obtain reliable pulse and HRV data even under motion or lighting variations.

3.2.1.3 Response Time

- Impact detection + decision time: ≤ 300 ms
- SOS alert transmission: ≤ 5 s after confirmation of a crash

Rationale: To protect and ensure the rider's safety, the urgency of the situation requires that emergency messages be sent and received promptly.

3.2.1.4 Lifespan and Maintenance

The Rakib helmet is expected to last about 3 years with normal use where each full charge lasts for around 8 hours and the battery will keep at least 80% of its capacity after 300 full charging cycles.

Rationale: These values are very suitable for delivery riders who use the helmet every day. As they make sure the system works for long hours without the need for frequent charging or replacing the sensors, all of this keeps maintenance low and performance reliable.

3.2.1.5 Communication Requirements

Use of 4G LTE for real-time data upload and SOS alert and uses GPS for live location tracking and route logging. WiFi/Bluetooth connections are provided as backup for short-range data synchronization.

- Uplink rate: ≥ 200 kbps
- GPS update interval: 2s in emergency mode

Rationale: The combination of LTE, GPS, and Bluetooth provides strong and stable communication in every environment since it allows fast data transfer and accurate tracking which ensures the helmet works reliably at all times.

3.2.2 Physical Characteristics

The Rakib Helmet is compliant with the ASTM F1446 safety standards, making sure that the impact absorption and resistance meets the thresholds needed to meet the standards [7]. The integrated electronic components are internally securely mounted and enclosed for the prevention of discomfort and to protect the helmet's structure.

Rationale: This design maintains the helmets safety and integrity while allowing the integration of sensors and modules without affecting comfort or weight balance.

3.2.3 Electrical Characteristics

Power Supply:

The system operates on a LiPo rechargeable battery that provides up to 8 hours of continuous power and supports the sensors and AI model.

Rationale: Battery used is a compact and lightweight model, making it suitable for wearable electronics and has enough runtime for daily use.

Power Consumption:

Maximum power draw is limited to 5W during operations to ensure safe thermal conditions and efficient power management.
Rationale: having a set limit of power controls the heat generation within the helmet, thus improving the users safety and extending the lifespan of the components.
Communication Interface:
Bluetooth is used for mobile connectivity and wireless data transmission between the helmet and the application.
Rationale: Provides low-energy operation and stable connectivity at a good data rate for transmitting sensor readings in real time.

3.2.4 Environmental Requirements

Operating Temperature: Components used can operate smoothly and reliably at temperatures from approximately -10 up to a 55 degrees celsius [6].
Rationale: This range takes into account Qatar's environmental conditions and protects the components from thermal stress.
Environmental Resistance: Electrical components are protected and enclosed within the helmet (under lining) , meeting IP54 standards to protect against dust and water splashes.
Rationale: Sensors are reliable in outdoor motorcycle environments and unstable weather conditions.
Shock and Vibration: The helmet shell is designed to theoretically withstand impacts up to 1500g without any internal component failure or critical damage [24].
Rationale: This prevents component detachment or data loss during severe impacts.

3.2.5 Safety and Fail-Safe Mechanisms

Impact Detection: SOS alert is activated automatically for impacts between 10–50g (critical crash) , with a buffer to reduce false alarms.
Rationale: Ensures emergency alerts are taken seriously and responded to promptly and filters out non-critical shocks .
System Diagnostics: Self-check to verify sensor connectivity, battery health, and data transmission are initiated when the system is started could also alert the user if something is not connected or not working.
Rationale: Automated diagnostics improve reliability and help detect hardware or communication faults.
Manual Override: The rider can cancel or reset alerts manually via the mobile app or the physical button on the helmet.
Rationale: Provides user control in case of false alarms or stable post crash conditions.

3.3 Software Subsystem

The software subsystem is embedded and run through a web server that manages the reception of sensor data , AI feedback, and communications within the mobile application. Its primary function is to enhance rider safety by detecting fatigue and crash conditions through intelligent data processing.

Data Acquisition and Processing: Sensor data from the PPG/HRV, accelerometer/gyroscope are collected, filtered, and processed locally before transmission to the app for application and logging.
Rationale: Local processing reduces latency and enables the detection to be accurate and fast acting in critical situations.

Machine Learning Integration: TensorFlow Lite models analyze data collected by the sensors and transmitted to the microcontroller to the web server. Then feedback and results are communicated back to the microcontroller for alerts and system response.
Rationale: AI models have no dependence on network connectivity and ensure immediate response during emergencies.

Data Storage and Communication: Firebase is used for storing logs and SOS records, accessible through the Rakib mobile application.
Rationale: Cloud based storage provides reliable data access and backup for long-term performance tracking.

Application Layer: The mobile app displays fatigue warnings, crash events, and battery status in real time. Allowing users to review ride data and change system settings.
*Rationale:*User interface ensures easy access to user friendly information and continuous performance monitoring.

3.4 Hardware Requirements

The Rakib AI helmet combines a variety of hardware components that work together to monitor and detect rider fatigue and crashes, ensuring safety, stability and reliability.

Main Control Unit: An ESP32 microcontroller with integrated Bluetooth and Wi-Fi networking powers the system and it acts as the main controller, gathering information from all sensors, controlling module-to-module communication, and connecting the helmet to the Rakib app for data processing and realtime feedback [19].

Sensors: To monitor fatigue, the helmet uses a PPG/HRV sensor which tracks heart rate variability and blood flow. To detect crashes a accelerometer/gyroscope sensor is used which tracks tilt, motion, and sudden impacts.

Audio & Alert system: For visual and audible feedback the helmet uses a buzzer, a speaker-microphone, and LED indicators. The LED indicators show the system status; blue indicates that fatigue is detected, , red indicates that the SOS message has not yet been sent, and green indicates that the SOS has been successfully sent. Before the SOS message is triggered, the buzzer serves as an extra warning by emitting an audible countdown. The speaker and microphone is used so the system delivers voice prompts and enables two-way communication through the Rakib mobile app for status updates or manual overrides.

Power Subsystem: This system is designed to work for 8 hours continuously one a single charge to achieve this a BMS, charger circuit, and fuel gauge support the rechargeable Li-Po battery that powers the helmet; this guarantees steady power distribution, safe charging, and uninterrupted performance for up to eight hours on a single charge.

3.5 Support Infrastructure

Technical Support:

If the users face any problems they can either contact a support email or online chat system for any technical or operational issues.

Service Continuity:

Firmware updates and AI model improvements are sent through the mobile app. This keeps the helmet always up to date, more reliable and better with every update

4. Support Requirements

Rakib Helmet requires very specific tools and conditions to ensure it continues working safely and effectively over time.

Mobile Interface:

A Bluetooth smartphone is required to pair with the helmet. The app will display the alerts, fatigue data and ride logs. The SOS works automatically, even without an app or Wifi connection.

Network and Cloud Connectivity:

The helmet uses Bluetooth 5.0 to connect with both the Firebase and mobile app to back up ride data securely. This allows users to check ride history and safety logs anytime they want.

Maintenance and Calibration:

It's recommended that every six months the users recalibrate the sensors and verify fatigue detection accuracy using the mobile app.

Software Updates and Data Security:

The system protects user privacy by applying AES-256 encryption for all the data transfers. It also supports updates called over-the-air (OTA) updates, this helps the firmware and AI models update automatically without the need to connect the helmet physically.

Technical Documentation:

Digital manuals and troubleshooting guides are available through the Rakib mobile app and also an official website to help assist users in setup and maintenance.

4.1 Challenges and Limitations

Technical Challenges:

Power management is one of the main challenges as the system must balance performance and battery life while ensuring accurate fatigue detection during motion.

AI and Data Challenges:

Because there are few real-world fatigue datasets for motorcycle riders [17], the AI model's accuracy might change at first, so continuous field testing is needed to make it more reliable.

Ensuring that the AI will work on the current ESP may be difficult however, two plans have been formulated in case, the first is to have a web server that houses the AI model and have back and forth communication with the microcontroller. And the second option is to use Raspberry pi that can accommodate both the sensors and AI model as it is a much powerful controller.

Operational Challenges:

Users must regularly charge the helmet and get used to fatigue alerts. Comfort and proper fit are also important for the system to function correctly.

4.2 Advantages

The Rakib Helmet helps prevent fatigue-related motorcycle accidents by warning the rider before he gets very tired [8]. In addition, it also provides automatic SOS alerts during crashes and stores ride data for analysis. All of these features work together to increase rider safety and help in improving safety systems.

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Appendix A Acronyms and Abbreviations

AI	Artificial Intelligence
BMS	Battery Management System
Esp32	Embedded System Platform (Microcontroller)
GPS	Global Positioning System
GUI	Graphical User Interference
HRV	Heart Rate Variability
IMU	Inertial Measurement Unit
LED	Light Emitting Diode
Li-Po	Lithium Polymer Battery
LTE	Long Term Evolution (4G Communication)
ML	Machine Learning
PPG	Photoplethysmography
SOS	Save Our Souls (Emergency Alert Signal)
WHO	World Health Organization

Appendix B Definition of Terms

Term	Definition
Microcontroller (ESP32)	The main processing unit that collects and processes sensor data, runs AI models, and manages communication.
PPG/HRV Sensor	Measures the rider's heart rate and variability to detect fatigue.
Accelerometer/Gyroscope Sensor	Detects head motion, tilt, and impact forces to identify crash or fatigue patterns.
Communication Module (LTE/GPS)	Sends SOS alerts with live GPS coordinates and uploads ride data to the cloud.
AI Fatigue Detection Model	A trained lightweight neural network that identifies early signs of rider drowsiness using heart rate and motion data.